



GRADE 9 NATURAL SCIENCES
MARCH TEST STUDY NOTES
SYSTEMS IN THE HUMAN BODY

1. MUSCULAR & SKELETAL SYSTEM

Pages: 50–51

The **musculoskeletal system** allows the body to move and provides support and protection for organs.

Main parts

- **Skeleton**
- **Muscles**

Functions of the skeleton

- Supports the body
- Protects internal organs
- Allows movement with muscles
- Produces blood cells
- Stores minerals such as calcium

Examples of important bones

Learners should recognise these bones:

- Skull
- Rib cage
- Vertebral column (spine)
- Femur (thigh bone)

2. CIRCULATORY SYSTEM

Pages: 56–59

The **circulatory system** transports substances throughout the body.

Functions

- Transport **oxygen** to cells
 - Transport **nutrients** to cells
 - Remove **carbon dioxide and other wastes**
-

Blood

Pages: 56–57

Blood transports substances around the body.

Components of blood

- **Red blood cells** – carry oxygen
 - **White blood cells** – fight infections
 - **Platelets** – help blood clot
 - **Plasma** – liquid part of blood that carries nutrients and wastes
-

Blood vessels

Page: 58

Blood moves through the body in **blood vessels**.

Types of blood vessels

Arteries

- Carry blood **away from the heart**

Veins

- Carry blood **towards the heart**

Capillaries

- Very small blood vessels
 - Exchange of **oxygen, nutrients and wastes** occurs here
-

Heart

 Pages: **58–59**

The **heart** is a muscular organ that **pumps blood throughout the body**.

Diagram practice: Heart

Learners should be able to **draw and label a simple heart diagram**.

Draw and label:

- Heart
- Arteries
- Veins
- Blood flow direction

Suggested drawing steps:

1. Draw a **large oval shape**.
 2. Divide the inside roughly into **four sections**.
 3. Draw **large blood vessels entering and leaving the heart**.
 4. Label the diagram clearly.
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3. RESPIRATORY SYSTEM

 Pages: **67–70**

The **respiratory system** allows the body to:

- Take in **oxygen**
 - Remove **carbon dioxide**
-

Anatomy of the respiratory system

 Pages: **68–69**

Learners only need to know the **basic functions** of these parts.

Parts and basic functions

Nasal cavity

- Air enters the body
- Air is filtered and warmed

Larynx

- Voice box
- Passage for air

Trachea

- Tube that carries air to the lungs

Bronchioles

- Smaller air tubes inside the lungs

Lungs

- Organs where gas exchange takes place

Alveoli

- Tiny air sacs where oxygen enters the blood

Diaphragm

- Muscle that helps breathing
-

Gas Exchange and Diffusion

 Page: **70**

Gas exchange occurs in the **alveoli**.

- **Oxygen diffuses** from the air in the lungs into the blood.
 - **Carbon dioxide diffuses** from the blood into the lungs.
 - Carbon dioxide is then **exhaled**.
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Diagram practice: Respiratory system

Learners should practise drawing and labelling:

- Nasal cavity
- Trachea
- Lungs

- Bronchioles
- Alveoli
- Diaphragm

Suggested drawing steps:

1. Draw the **nose and airway tube**.
2. Split the airway into **two lungs**.
3. Draw small **branches inside the lungs (bronchioles)**.
4. Label all parts clearly.

4. EXCRETORY SYSTEM

 Pages: **77–79**

The **excretory system** removes wastes from the body and helps maintain the body's internal balance.

Main organs (must know for diagram)

- Kidneys
- Ureters
- Bladder
- Urethra

Functions of the kidneys

 Pages: **78–79**

Learners must **only know the names** of these functions:

- **Homeostasis**
- **Filtration**
- **Reabsorption**
- **Maintain pH levels**
- **Regulate electrolyte–water balance**

No detailed explanations are required.

Diagram practice: Excretory system

Learners should be able to draw a **simple labelled diagram**.

Draw and label:

- Kidneys
- Ureters
- Bladder
- Urethra

Suggested drawing steps:

1. Draw **two bean-shaped kidneys**.
2. Draw **two thin tubes (ureters)** going downwards.
3. Draw the **bladder** at the bottom.
4. Draw the **urethra** leaving the bladder.

5. THE FIVE SENSES

Learners must know the **sense and the organ responsible**.

Sense Organ

Sight Eyes

Hearing Ears

Smell Nose

Taste Tongue

Touch Skin

Important:

Only the **sense and the organ** must be known.

No functions of different parts are required.